KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

YEAR 10 TRANSITION

EXTENSION BOOKLET

FOR FAST FINISHERS · CHALLENGE QUESTIONS

CHAPTER 2

YEAR 10 MATHEMATICS · CHAPTER 2 · EXTENSION: FAST FINISHERS

Measurement

One page of harder questions for each exercise in Chapter 2. Finish the exercise in your Chapter 2 booklet first, then come here. Worked answers are at the back.

EXERCISE	EXTENSION TOPIC	PAGE	DONE
2A	Converting units of measurement	2	☐
2B	Accuracy of measuring instruments	3	☐
2C	Very small and very large units	4	☐
2D	Perimeter	5	☐
2E	Circumference	6	☐
2F	Area	7	☐
2G	Area of circles and sectors	8	☐
2H	Composite shapes	9	☐
2I	Surface area of prisms	10	☐
2J	Volume of prisms and cylinders	11	☐

EVERY CARD HAS A TYPE

- Understanding
- Reasoning
- Problem solving

MIXED IN

Review 2A questions from earlier exercises

NO DIAGRAM?

✎ Draw a diagram first, in the dotted space

CHALLENGE

★☆☆ to ★★★

The last card on each page is the challenge.

NAME _____ CLASS _____ TEACHER _____

Artwork: the “elephants” of Elephant Valley, part of the Mandelbrot set, coloured gold.

Converting units of measurement

FAST FINISHERS

EXTENSION Finished exercise 2A? Try these. Show your working in each box. 1 L = 1000 mL and 1 cm³ holds 1 mL.

Understanding · Reasoning · Problem solving

1 Understanding ★☆☆

Convert:

a 3.6 km to m

b 450 mm to m

c 2.5 L to mL

d 7200 g to kg

a _____ b _____ c _____ d _____

2 Understanding ★☆☆

Convert:

a 3 m² to cm²

b 45 000 mm² to cm²

c 2 m³ to cm³

a _____ b _____ c _____

3 Reasoning ✎ Draw a diagram ★★☆

Find the error. Ben says, "5 m² = 500 cm², because there are 100 cm in a metre."

a Draw a 1 m by 1 m square and label its sides in cm. Use it to explain Ben's mistake.

b Find the correct answer.

b _____

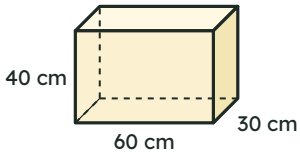
4 Problem solving ★★☆

A fish tank is a rectangular prism.

a Find its volume in cm³.

b Find its capacity in litres.

c How many 9 L buckets of water will fill it?



a _____ b _____ c _____

5 Problem solving ★★☆

A running track is 400 m around.

a How many laps make 5 km?

b Jess runs 3.2 km every day for 2 weeks. How far is this in km and in m?

a _____ b _____

6 Reasoning ★★★

A rectangular paddock is 0.8 km long and 350 m wide. 1 hectare (ha) = 10 000 m².

a Find its area in m².

b Find its area in hectares.

c Find its area in km².

d Explain why you must change both lengths to the same unit first.

a _____ b _____ c _____

Accuracy of measuring instruments

FAST FINISHERS

EXTENSION Finished exercise 2B? Try these. Show your working in each box. Limits of accuracy: ± 0.5 × the smallest unit of measurement.

Understanding · Reasoning · Problem solving

1 Understanding ★☆☆

Give the lower and upper limits for each measurement.

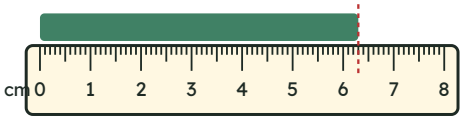
a 8 cm, measured to the nearest cm

b 3.6 kg, to the nearest 0.1 kg

c 250 mL, to the nearest 10 mL

a _____ b _____ c _____

2 Understanding ★☆☆



a Read the length of the green bar, to the nearest millimetre.

b What are the limits of accuracy of this reading?

a _____ b _____

3 Reasoning ★★☆

A rope is 12 m long, measured to the nearest metre.

a Explain why the rope cannot be 12.6 m long.

b Could the rope be 11.5 m long? Explain.

c Write the range of possible lengths.

c _____

4 Problem solving ✎ Draw a diagram ★★☆

A rectangle is measured as 7 cm by 4 cm, both to the nearest cm.

a Write the limits for each side.

b Find the smallest possible perimeter.

c Find the largest possible perimeter.

b _____ c _____

5 Problem solving Review 2A ★★☆

A bag of flour is labelled 2.5 kg, to the nearest 0.1 kg.

a Write its limits of accuracy in grams.

b A recipe uses 300 g. Can you always make 8 batches from one bag? Explain.

a _____

6 Reasoning ★★★

A square tile has a side of 10 cm, measured to the nearest cm.

a Find the smallest and largest possible area.

b How far could the true area be from 100 cm²?

c The side could be out by 5%. Explain why the area could be out by about 10%.

a _____ b _____

Very small and very large units

FAST FINISHERS

EXTENSION Finished exercise 2C? Try these. Show your working in each box.
Scientific notation: $a \times 10^m$, where $1 \le a < 10$.

Understanding · Reasoning · Problem solving

1 Understanding ★☆☆

Write in scientific notation.

a 45 000

b 0.0072

c 3 080 000

a _____ b _____ c _____

2 Understanding ★☆☆

a Write 6.1×10^5 as a basic numeral.

b Write 2.4×10^{-4} as a basic numeral.

c Which is larger: 9.9×10^3 or 1.1×10^4? Explain.

a _____ b _____ c _____

3 Reasoning ★★☆

a Round 0.004 862 to 2 significant figures.

b Round 38 470 to 3 significant figures.

c How many significant figures are in 0.0305? Explain which zeros count.

a _____ b _____ c _____

4 Problem solving Review 2A ★★☆

a How many seconds are in 1 day? Write your answer in scientific notation.

b How many seconds are in a year of 365 days? Give your answer in scientific notation to 3 significant figures.

c 1 millisecond (ms) = 0.001 s. How many ms are in one minute?

a _____ b _____ c _____

5 Understanding ★★☆

1 KB = 1000 bytes, 1 MB = 1000 KB, 1 GB = 1000 MB.

a A phone has 64 GB of storage. How many 4.5 MB photos fit?

b Write 64 GB in bytes, in scientific notation.

a _____ b _____

6 Reasoning ★★★

A human hair is about 8×10^{-5} m wide. A sheet of paper is about 1×10^{-4} m thick.

a Which is thicker? Explain.

b How many hairs side by side would make 1 cm?

c The Sun is 1.5×10^8 km away. Light travels 3×10^5 km each second. How long does sunlight take to reach us?

a _____ b _____ c _____

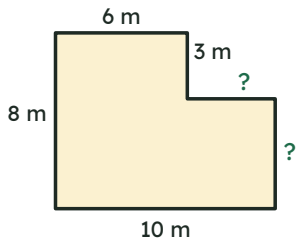
Perimeter

FAST FINISHERS

EXTENSION Finished exercise 2D? Try these. Show your working in each box. Mark any missing lengths on the diagram before you add. Understanding · Reasoning · Problem solving

1 Understanding ★☆☆

Find the missing lengths marked ?, then find the perimeter of the shape.



? _____ ? _____ P = _____

2 Understanding · Draw a diagram ★☆☆

Work backwards. A rectangle has a perimeter of 46 cm and a length of 15 cm.

a Find its width.

b A square has the same perimeter. Find its side length.

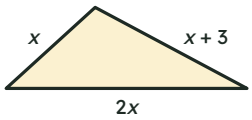
a _____ b _____

3 Reasoning ★★☆

a Write an expression for the perimeter.

b The perimeter is 31 cm. Find x.

c Find the length of each side.



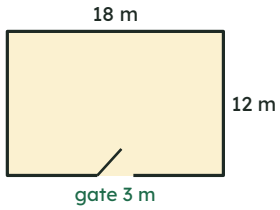
a _____ b _____ c _____

4 Problem solving ★★☆

A yard is fenced on all four sides, leaving a 3 m gap for a gate. Fencing costs $45 per metre.

a Find the length of fencing needed.

b Find the cost.



a _____ b _____

5 Problem solving · Review 2A · Draw a diagram ★★☆

A rectangular garden is 3.6 m by 240 cm. Edging is sold in 1.2 m lengths at $8.50 each.

a Find the perimeter in metres.

b How many lengths of edging are needed?

c Find the cost.

a _____ b _____ c _____

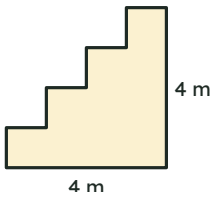
6 Reasoning ★★★

Each step is 1 m wide and 1 m high.

a Find the perimeter of the shape.

b Explain a quick way to get the answer without adding every side.

c Would the perimeter change with 8 steps each 0.5 m? Explain.



a _____

Circumference

FAST FINISHERS

EXTENSION Finished exercise 2E? Try these. Show your working in each box. Use $C = \pi d$ or $C = 2\pi r$. Round to **2 decimal places**.

Understanding · Reasoning · Problem solving

1 Understanding · ✎ Draw a diagram ★☆☆

Find the circumference of a circle with:

a diameter 12 cm

b radius 4.5 m

a _____ b _____

2 Understanding · ✎ Draw a diagram ★☆☆

Work backwards. A circle has a circumference of 50 cm.

a Find its diameter.

b Find its radius.

a _____ b _____

3 Reasoning ★★☆

A bike wheel has a diameter of 66 cm.

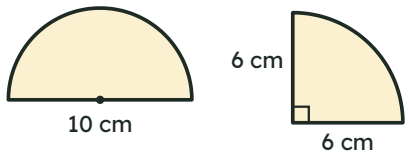
a Find its circumference.

b How many full turns does the wheel make in 1 km?

c Explain why you must change 1 km to cm first.

a _____ b _____

4 Understanding ★★☆



Find the perimeter of each shape. Include the straight edges.

semicircle _____ **quarter circle** _____

5 Problem solving · Review 2B ★★☆

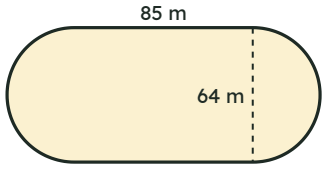
A coin has a diameter of 2.5 cm, measured to the nearest mm.

a Write the limits of accuracy for the diameter.

b Find the smallest and largest possible circumference.

a _____ b _____

6 Problem solving ★★★



A track has two 85 m straights and two semicircular ends.

a Find the length of one lap.

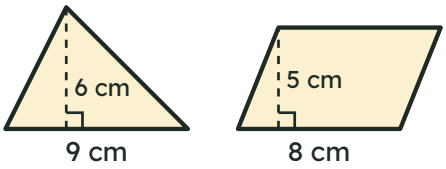
b How many laps must Ivy run to cover at least 3 km?

a _____ b _____

EXTENSION Finished exercise 2F? Try these. Show your working in each box. Write the formula first, then substitute.

Understanding · Reasoning · Problem solving

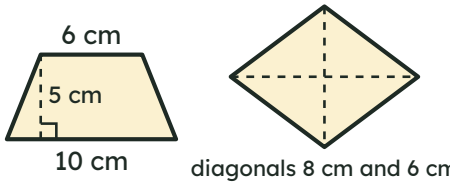
1 Understanding ★☆☆



Find the area of the triangle and the parallelogram.

triangle _____ **parallelogram** _____

2 Understanding ★☆☆



Find the area of the trapezium and the rhombus.

trapezium _____ **rhombus** _____

3 Reasoning · ✎ Draw a diagram ★★☆

Work backwards.

a A triangle has an area of 36 cm² and a base of 9 cm. Find its height.

b A rectangle has an area of 84 m² and a length of 12 m. Find its width.

a _____ **b** _____

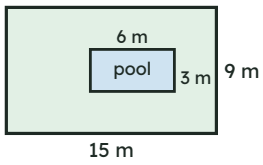
4 Problem solving · Review 2D ★★☆

A backyard has a pool. The rest is covered in turf costing $12.50 per m².

a Find the area of turf.

b Find the cost of the turf.

c Find the perimeter of the pool.



a _____ **b** _____ **c** _____

5 Reasoning · ✎ Draw a diagram ★★☆

Find the error. Ella says the area of a triangle with base 10 cm and height 7 cm is 70 cm².

a Draw the triangle inside a 10 cm by 7 cm rectangle and use it to explain her mistake.

b Find the correct area.

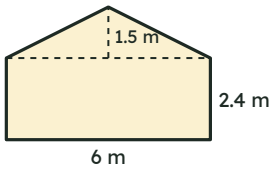
b _____

6 Problem solving ★★★

The end wall of a shed is a rectangle with a triangle on top. It needs two coats of paint. 1 L covers 12 m².

a Find the area of the wall.

b Paint is sold in 1 L tins. How many tins are needed?



a _____ **b** _____

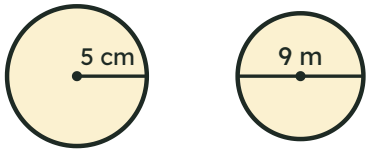
Area of circles and sectors

FAST FINISHERS

EXTENSION Finished exercise 2G? Try these. Show your working in each box. Sector area = $\frac{\theta}{360} \times \pi r^2$. Round to **2 decimal places**.

Understanding · Reasoning · Problem solving

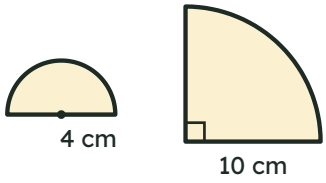
1 Understanding ★☆☆



Find the area of each circle.

left _____ **right** _____

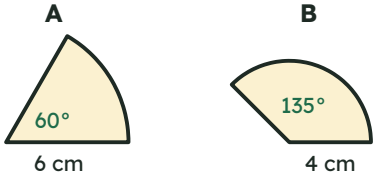
2 Understanding ★☆☆



Find the area of the semicircle and the quarter circle.

semicircle _____ **quarter** _____

3 Reasoning ★★☆



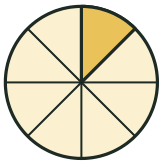
a Find the area of each sector.
b Explain how a smaller radius can give the same area.

A _____ **B** _____

4 Problem solving ★★☆

A pizza is cut into 8 equal slices.

a Find the angle of each slice.
b Find the area of one slice.
c Find the area of 3 slices.



a _____ **b** _____ **c** _____

5 Reasoning · Review 2E · ✎ Draw a diagram ★★☆

Work backwards. A circle has a circumference of 37.70 cm.

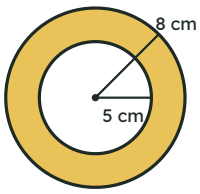
a Find its radius.
b Find its area.

a _____ **b** _____

6 Problem solving ★★★

A metal washer is a ring shape.

a Find the area of the outer circle.
b Find the area of the hole.
c Find the area of the shaded ring.
d Tom says the ring area is $\pi \times 3^2$. Explain why he is wrong.



a _____ **b** _____ **c** _____

Composite shapes

FAST FINISHERS

EXTENSION Finished exercise 2H? Try these. Show your working in each box. Split into shapes you know. Round to **2 decimal places**.

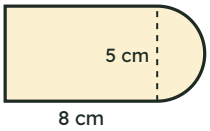
Understanding · Reasoning · Problem solving

1 Understanding ★☆☆

A rectangle has a semicircle on one end.

a Find its area.

b Find its perimeter.



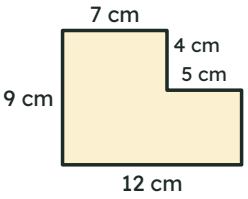
a _____ b _____

2 Reasoning ★☆☆

a Find the area by **adding** two rectangles.

b Find the area by **subtracting** a rectangle.

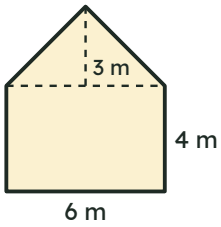
c Which method do you prefer? Why?



a _____ b _____

3 Understanding ★★☆

Find the area of the front of this cubby house.



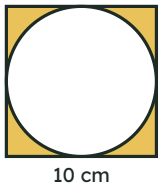
A = _____

4 Problem solving ★★☆

A circle fits exactly inside a square. The corners are shaded.

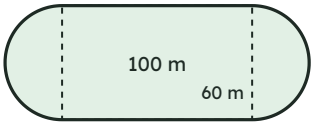
a Find the shaded area.

b What percentage of the square is shaded?



a _____ b _____

5 Problem solving · Review 2E ★★☆



A sports field is a rectangle with a semicircle on each end.

a Find its area.

b Find its perimeter.

a _____ b _____

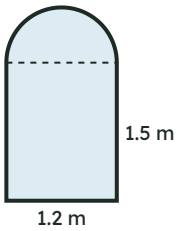
6 Problem solving ★★★

A window is a rectangle with a semicircle on top.

a Find the area of glass.

b Find the length of the outside frame (not the dashed line).

c The frame costs $35 per metre. Find its cost.



a _____ b _____ c _____

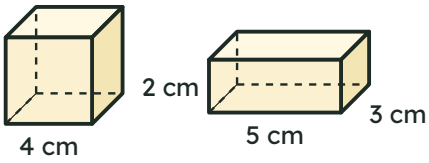
Surface area of prisms

FAST FINISHERS

EXTENSION Finished exercise 2I? Try these. Show your working in each box. Surface area = total area of all faces. Sketch the net.

Understanding · Reasoning · Problem solving

1 Understanding ★☆☆



Find the surface area of the cube and the rectangular prism.

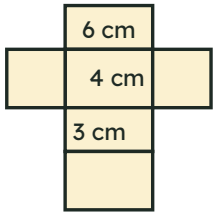
cube _____ prism _____

2 Understanding ★☆☆

This is the net of a rectangular prism.

a What are the length, width and height of the prism?

b Find its surface area.



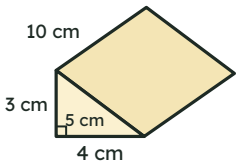
a _____ b _____

3 Reasoning Review 2F ★★☆

This prism has right-angled triangles at each end.

a Find the area of one triangle.

b Find the surface area of the prism.



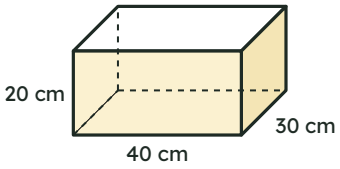
a _____ b _____

4 Problem solving ★★☆

A storage box has **no lid**.

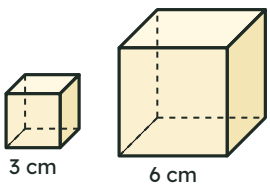
a Find the area of cardboard needed in cm².

b Change your answer to m².



a _____ b _____

5 Reasoning ★★☆



a Find the surface area of each cube.

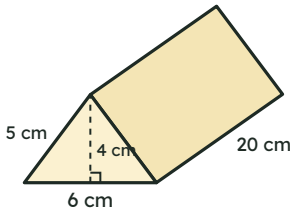
b The side doubled. What happened to the surface area? Explain why.

a _____ b _____

6 Problem solving ★★★

A chocolate box is a triangular prism. Each end is an isosceles triangle.

a Use Pythagoras' theorem to check the sloping sides are 5 cm.



b Find the surface area of the box.

a _____ b _____

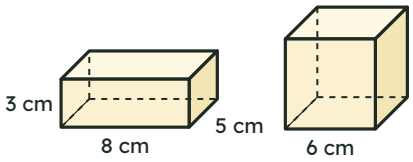
Volume of prisms and cylinders

FAST FINISHERS

EXTENSION Finished exercise 2J? Try these. Show your working in each box. $V = Ah$ and 1 cm³ = 1 mL. Round to **2 decimal places**.

Understanding · Reasoning · Problem solving

1 Understanding ★☆☆

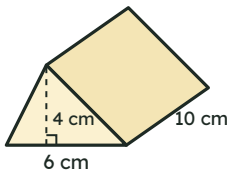


Find the volume of the rectangular prism and the cube.

prism _____ cube _____

2 Understanding ★☆☆

Find the volume of this triangular prism.



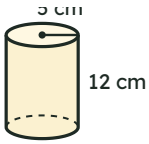
V = _____

3 Understanding ★★☆

A tin is a cylinder.

a Find its volume.

b How many litres does it hold?



a _____ b _____

4 Reasoning · ✎ Draw a diagram ★★☆

Work backwards.

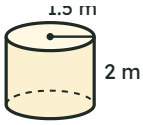
a A rectangular prism has a volume of 360 cm³ and a base 9 cm by 8 cm. Find its height.

b A cylinder with radius 5 cm holds exactly 1 L. Find its height.

a _____ b _____

5 Problem solving ★★☆

A rainwater tank is a cylinder with diameter 3 m and height 2 m.



a Find its volume in m³.

b How many litres does it hold? (1 m³ = 1000 L)

c A family uses 450 L a day. How many full days will a full tank last?

a _____ b _____ c _____

6 Problem solving · Review 2A · ✎ Draw a diagram ★★★

A concrete path is 12 m long, 1.2 m wide and 10 cm deep. Concrete costs $230 per m³.

a Find the volume of concrete in m³.

b Find the cost.

c Concrete is only sold in multiples of 0.2 m³. How much must be ordered, and what will it cost?

a _____ b _____ c _____

ANSWERS Worked answers. Check your method as well as your answer. Rounded answers may differ slightly in the last decimal place.

2A · Converting units of measurement

1. **a 3600 m b 0.45 m c 2500 mL d 7.2 kg**
2. **a** $3 \times 100^2 =$ **30 000 cm²** **b** $45\,000 \div 10^2 =$ **450 cm²** **c** $2 \times 100^3 =$ **2 000 000 cm³**
3. **a** A 1 m square is 100 cm × 100 cm = 10 000 cm², so for area you multiply by 100^2 not 100. **b** $5 \times 10\,000 =$ **50 000 cm²**
4. **a** $60 \times 30 \times 40 =$ **72 000 cm³** **b** 72 000 mL = **72 L** **c** $72 \div 9 =$ **8 buckets**
5. **a** $5000 \div 400 =$ **12.5 laps** **b** $3.2 \times 14 =$ **44.8 km = 44 800 m**
6. **a** $800 \times 350 =$ **280 000 m²** **b** $\div 10\,000 =$ **28 ha** **c** $0.8 \times 0.35 =$ **0.28 km²** **d** $0.8 \times 350 = 280$ mixes km and m and gives a meaningless number.

2B · Accuracy of measuring instruments

1. **a 7.5 cm – 8.5 cm b 3.55 kg – 3.65 kg c 245 mL – 255 mL**
2. **a 6.3 cm** (63 mm) **b 6.25 cm – 6.35 cm**
3. **a** 12.6 m rounds to 13 m, not 12 m. **b** Yes. 11.5 m is the lower limit; it rounds up to 12 m. **c 11.5 m ≤ length < 12.5 m**
4. **a** 6.5–7.5 cm and 3.5–4.5 cm **b** $2(6.5 + 3.5) =$ **20 cm** **c** $2(7.5 + 4.5) =$ **24 cm**
5. **a 2450 g – 2550 g** **b** $8 \times 300 = 2400$ g, which is less than the smallest possible amount (2450 g), so **yes**.
6. **a** $9.5^2 =$ **90.25 cm²**, $10.5^2 =$ **110.25 cm²** **b** up to **10.25 cm²** **c** Area = side × side, so the error happens twice ($1.05 \times 1.05 \approx 1.10$).

2C · Very small and very large units

1. **a** 4.5×10^4 **b** 7.2×10^{-3} **c** 3.08×10^6
2. **a 610 000 b 0.000 24 c** $1.1 \times 10^4 = 11\,000$, which is more than 9900. Compare the powers of 10 first.
3. **a 0.0049 b 38 500 c 3**. The leading zeros don't count; the zero between 3 and 5 does.
4. **a** $24 \times 60 \times 60 = 86\,400 =$ **8.64×10^4** **b** $31\,536\,000 \approx$ **3.15×10^7** **c** $60 \div 0.001 =$ **60 000 ms**
5. **a** $64\,000 \div 4.5 = 14\,222.2$, so **14 222 photos** **b** $64 \times 10^9 =$ **6.4×10^{10} bytes**
6. **a The paper**: 0.0001 m > 0.000 08 m **b** $0.01 \div 0.000\,08 =$ **125 hairs** **c** $1.5 \times 10^8 \div (3 \times 10^5) =$ **500 s = 8 min 20 s**

2D · Perimeter

1. Missing: $10 - 6 =$ **4 m** and $8 - 3 =$ **5 m**. P = 10 + 8 + 6 + 3 + 4 + 5 = **36 m**
2. **a** $46 \div 2 = 23$; $23 - 15 =$ **8 cm** **b** $46 \div 4 =$ **11.5 cm**
3. **a** $x + x + 3 + 2x =$ **$4x + 3$** **b** $4x = 28$, **$x = 7$** **c 7 cm, 10 cm, 14 cm**
4. **a** $2(18 + 12) - 3 =$ **57 m** **b** $57 \times 45 =$ **$2565**
5. **a** 240 cm = 2.4 m; $2(3.6 + 2.4) =$ **12 m** **b** $12 \div 1.2 =$ **10** **c** $10 \times 8.50 =$ **$85**
6. **a** $4 + 4 + 8 \times 1 =$ **16 m** **b** The vertical steps add to 4 m and the horizontal steps add to 4 m, so P is the same as a 4 m × 4 m square. **c No**, still 16 m: the steps still add to 4 m each way.

2E · Circumference

1. **a** $\pi \times 12 =$ **37.70 cm** **b** $2 \times \pi \times 4.5 =$ **28.27 m**
2. **a** $50 \div \pi =$ **15.92 cm** **b** $15.92 \div 2 =$ **7.96 cm**
3. **a** $\pi \times 66 =$ **207.35 cm** **b** $100\,000 \div 207.35 = 482.3$, so **482 full turns** **c** The circumference is in cm, so both distances must be in the same unit.
4. Semicircle: $\pi \times 10 \div 2 + 10 =$ **25.71 cm**. Quarter circle: $2 \times \pi \times 6 \div 4 + 6 + 6 =$ **21.42 cm**
5. **a 2.45 cm – 2.55 cm** **b** $\pi \times 2.45 =$ **7.70 cm**; $\pi \times 2.55 =$ **8.01 cm**
6. **a** $2 \times 85 + \pi \times 64 = 170 + 201.06 =$ **371.06 m** **b** $3000 \div 371.06 = 8.08$, so **9 laps**

ANSWERS Worked answers. Check your method as well as your answer. Rounded answers may differ slightly in the last decimal place.

2F · Area

1. Triangle: ½ × 9 × 6 = **27 cm²**. Parallelogram: 8 × 5 = **40 cm²**
2. Trapezium: ½ × 5 × (6 + 10) = **40 cm²**. Rhombus: ½ × 8 × 6 = **24 cm²**
3. **a** ½ × 9 × h = 36, so h = 72 ÷ 9 = **8 cm** **b** 84 ÷ 12 = **7 m**
4. **a** 15 × 9 − 6 × 3 = 135 − 18 = **117 m²** **b** 117 × 12.50 = **$1462.50** **c** 2(6 + 3) = **18 m**
5. **a** The triangle is half of the rectangle, so you must multiply by ½. **b** ½ × 10 × 7 = **35 cm²**
6. **a** 6 × 2.4 + ½ × 6 × 1.5 = 14.4 + 4.5 = **18.9 m²** **b** 2 × 18.9 = 37.8 m²; 37.8 ÷ 12 = 3.15 L, so **4 tins**

2G · Area of circles and sectors

1. Left: π × 5² = **78.54 cm²**. Right: r = 4.5, π × 4.5² = **63.62 m²**
2. Semicircle: ½ × π × 4² = **25.13 cm²**. Quarter circle: ¼ × π × 10² = **78.54 cm²**
3. **a** A: 60/360 × π × 6² = **18.85 cm²**; B: 135/360 × π × 4² = **18.85 cm²** (both 6π) **b** B has a smaller radius but a bigger angle, so it is a bigger fraction of its circle.
4. **a** 360 ÷ 8 = **45°** **b** π × 15² ÷ 8 = **88.36 cm²** **c** 3 × 88.357 = **265.07 cm²**
5. **a** 37.70 ÷ (2π) = **6 cm** **b** π × 6² = **113.10 cm²**
6. **a** π × 8² = **201.06 cm²** **b** π × 5² = **78.54 cm²** **c** **122.52 cm²** **d** You subtract the areas (64π − 25π = 39π), not the radii first. π × 3² = 28.27 cm² is far too small.

2H · Composite shapes

1. **a** 8 × 5 + ½ × π × 2.5² = 40 + 9.82 = **49.82 cm²** **b** 8 + 8 + 5 + ½ × π × 5 = 21 + 7.85 = **28.85 cm**
2. **a** 12 × 5 + 7 × 4 = 60 + 28 = **88 cm²** **b** 12 × 9 − 5 × 4 = 108 − 20 = **88 cm²** **c** Either, with a reason.
3. 6 × 4 + ½ × 6 × 3 = 24 + 9 = **33 m²**
4. **a** 10² − π × 5² = 100 − 78.54 = **21.46 cm²** **b** 21.46 ÷ 100 = **21.5%**
5. **a** 100 × 60 + π × 30² = 6000 + 2827.43 = **8827.43 m²** **b** 2 × 100 + π × 60 = **388.50 m**
6. **a** 1.2 × 1.5 + ½ × π × 0.6² = 1.8 + 0.57 = **2.37 m²** **b** 1.5 + 1.5 + 1.2 + ½ × π × 1.2 = **6.08 m** **c** 6.085 × 35 = **$212.97** (or $212.80 using 6.08 m)

2I · Surface area of prisms

1. Cube: 6 × 4² = **96 cm²**. Prism: 2(5 × 3 + 5 × 2 + 3 × 2) = 2 × 31 = **62 cm²**
2. **a** **6 cm × 4 cm × 3 cm** **b** 2(24 + 18 + 12) = **108 cm²**
3. **a** ½ × 4 × 3 = **6 cm²** **b** 2 × 6 + 10 × (3 + 4 + 5) = 12 + 120 = **132 cm²**
4. **a** 40 × 30 + 2 × 40 × 20 + 2 × 30 × 20 = 1200 + 1600 + 1200 = **4000 cm²** **b** ÷ 10 000 = **0.4 m²**
5. **a** 6 × 3² = **54 cm²**; 6 × 6² = **216 cm²** **b** It is **4 times** bigger: each face is 2 × 2 = 4 times the area.
6. **a** Half the base is 3 cm: √(3² + 4²) = √25 = **5 cm** ✓ **b** 2 × ½ × 6 × 4 + 20 × (5 + 5 + 6) = 24 + 320 = **344 cm²**

2J · Volume of prisms and cylinders

1. Prism: 8 × 5 × 3 = **120 cm³**. Cube: 6³ = **216 cm³**
2. A = ½ × 6 × 4 = 12 cm²; V = 12 × 10 = **120 cm³**
3. **a** π × 5² × 12 = **942.48 cm³** **b** 942.48 mL = **0.94 L**
4. **a** 360 ÷ (9 × 8) = **5 cm** **b** 1 L = 1000 cm³; 1000 ÷ (π × 5²) = **12.73 cm**
5. **a** π × 1.5² × 2 = **14.14 m³** **b** **14 137 L** **c** 14 137 ÷ 450 = 31.4, so **31 days**
6. **a** 12 × 1.2 × 0.1 = **1.44 m³** **b** 1.44 × 230 = **$331.20** **c** **1.6 m³**, costing 1.6 × 230 = **$368**

ABOUT THE ARTWORK

The pictures in this booklet are parts of the Mandelbrot set, drawn by a computer that repeats one simple rule, z → z² + c, hundreds of times for every point. This one is Elephant Valley, where curling shapes like elephant trunks line up along the edge of the set. Zoom in on any part of the edge and you find more detail, forever. Maths can make beautiful things.

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We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.